

The Garage Separation and Sprinklers

The wall between the garage and the house is a fire barrier. Every number here is about how long it holds.

A garage holds a car, fuel, paint and tools — the most likely place in the house for a fire to start, and the place where nobody is asleep. The separation exists to buy time for the people who are.

The numbers here

Garage side of walls separating garage from residence and attic	1/2 in gypsum board
Garage ceiling with habitable rooms above	5/8 in Type X gypsum
Type X fire-retardant drywall	5/8 in
Solid wood garage-to-house door — minimum thickness	1 3/8 in

Half-inch ordinary board on the walls, five-eighths Type X on a ceiling with rooms above it. The ceiling gets the better material because fire goes up, and what is above it is someone's bedroom floor. The structure supporting that separation — the posts and beams — must be protected to the same standard.

The door

Acceptable garage-to-house doors	Four kinds: solid wood at least 1 3/8 inches thick, solid or honeycomb-core steel at least 1 3/8 inches, a 20-minute fire-rated door, or an equivalent listed assembly.
Required features	Self-closing and self-latching. A fire door propped or left ajar is a hole in the wall.
Absolutely prohibited	An opening from a garage directly into a sleeping room. There is no door rating that makes it acceptable.
The egress path	The pathway to the primary egress door out of the house cannot be through the garage.

Residential sprinklers

A sprinkler head is a heat-sensitive plug holding back water. Four things stop one working or distributing water properly: paint on the head, a stored item or shelf obstructing the spray pattern, corrosion, and physical damage to the head or its deflector.

Paint is the one asked about specifically: paint insulates the heat-sensitive element so it will not trigger at the right temperature, and the sprinkler stays shut through the fire it was fitted to fight. A painted head is always a defect, however neat the paintwork.

Say it back

Cover the page. Answer each one out loud.

1. Gypsum thickness on the garage wall, and on a ceiling with rooms above?
2. Why does the ceiling get the better material?
3. Name the four acceptable garage-to-house doors.
4. What two operating features must that door have?
5. Which opening from a garage is absolutely prohibited?
6. Can the primary egress path run through the garage?
7. Name four things that stop a sprinkler head working.
8. Why specifically is a painted head a defect?